THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 1–4: **Large Numbers, Arithmetic Expressions, A Peek Beyond the Point, Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. If $n = 10$, then $\frac{n(n+1)}{2}$ equals: 1

(a) 54 **(b)** 55 **(c)** 57 **(d)** 56

2. The smallest number that must be added to 34,56,789 to get the next lakh is: 1

(a) 43,111 **(b)** 99,999 **(c)** 43,211 **(d)** 44,211

3. 12 × 38 + 12 × 62 is equal to 12 × ?. The missing number is: 1

(a) 100 **(b)** 101 **(c)** 99 **(d)** 112

4. Which of these is closest to 1? 1

(a) 0.89 **(b)** 1.2 **(c)** 0.95 **(d)** 1.08

5. Which expression is always even for a whole number n? 1

(a) $2n + 4$ **(b)** $n + 2$ **(c)** $2n + 1$ **(d)** $3n$

6. If 1 lakh = 1,00,000, how many lakhs are there in 1,25,00,000? 1

(a) 124 **(b)** 125 **(c)** 135 **(d)** 126

7. A cyclist covers 30 km in the first hour and 2 km less every following hour for 5 hours in all. The total distance (km) is: 1

(a) 129 **(b)** 130 **(c)** 135 **(d)** 131

8. What must be added to 2.35 to get 5? 1

(a) 2.75 **(b)** 2.65 **(c)** 2.55 **(d)** 3.65

9. If $x = 4$ and $y = 5$, the value of $3x - 2y + xy$ is: 1

(a) 21 **(b)** 22 **(c)** 24 **(d)** 23

10. The number of digits in the product 1,25,000 × 8,000 is: 1

(a) 11 **(b)** 12 **(c)** 9 **(d)** 10

11. The value of $(3 \times (18 - 4) + 6) \div (2 \times (7 + 1))$ is: 1

(a) 5 **(b)** 197 **(c)** 72 **(d)** 3

12. Which fraction is equal to 0.125? 1

(a) $\frac{1}{8}$ **(b)** $\frac{1}{12}$ **(c)** $\frac{1}{125}$ **(d)** $\frac{1}{4}$

13. **Assertion (A):** 'Five more than three times n' is written as 3n + 5. 1
Reason (R): Three times n is written as 3n.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** 1 million = 10 lakh. 1
Reason (R): 1 billion = 1000 million.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** $(3 + 4) \times 2 = 14$. 1
Reason (R): $3 + 4 \times 2 = 11$.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Use the distributive property to find 37 × 48 + 37 × 52. 2

17. Using each of the digits $2, 5, 8, 4$ exactly once, form the largest and the smallest numbers having two digits before and two digits after the decimal point. Find their difference. 2

18. Write the sum of three consecutive multiples of 2, starting with $2n$, and simplify it. 2

19. A number is 1 lakh more than 1,23,45,678. Write the number and the number which is 1 crore less than it (if possible). 2

20. Without finding the products, compare 15 × 10 and 14 × 11. Check by computing. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Write the sum of 3 consecutive numbers starting at n as an expression in n, and evaluate it for $n = 7$. 3

22. The speed of light is about 3,00,000 km per second. How far does it travel in 8 seconds? Write the answer in words. 3

23. A rectangular garden is 20 m long and 12 m wide. A path of width 2 m runs around outside. Write an expression for the area of the path and evaluate it. 3

24. 2.5 kg of rice costs ₹135. Find the price of 1 kg and the cost of 4.5 kg. 3

OR

Taxi A charges ₹20 plus ₹15 per km and taxi B charges ₹60 plus ₹10 per km. Write both fares for k km. For a journey of 6 km, which taxi is cheaper and by how much? 3

25. A bank gives ₹1,25,000 loan to each of 96 persons. Find the total, write it in words and round it to the nearest crore. 3

OR

A shop sells 30 notebooks at ₹40 each and 25 pens at ₹15 each, and gives a discount of ₹25 on every 5 notebooks. Write an expression for the total amount and find it (discount counted once for each full set of 5 notebooks). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Ramesh buys 3 notebooks at ₹25 each and 2 pens at ₹10 each. He pays with a ₹100 note. 5

(i) Write one expression for the total cost.

(ii) Find its value.

(iii) How much change does he get?

(iv) Next day he buys 5 notebooks and 2 pens and pays with ₹200. Write an expression for the cost and find the change.

27. A shopkeeper mixes 1.5 kg of tea costing ₹120 per kg with 2.5 kg of tea costing ₹200 per kg. Find the total cost, the total weight and the cost of 1 kg of the mixture. 5

OR

Priya has ₹x. Her brother has ₹15 more than twice her money. Write expressions for the brother's money and for their total money, and find both when $x = 40$. 5

END OF QUESTION PAPER — Total: 50 Marks

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